SECR2043 OPERATING SYSTEMS

Name	: Chau Ying Jia	Marks
Student ID	: A23CS0213	
Section	: 02	

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write "**No output**" instead.

<table>
<tr><td rowspan="3">1.</td><td colspan="2">Display your current working directory.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>pwd</td><td>chau@secr2043:~$ pwd
/home/chau</td></tr>
<tr><td rowspan="3">2.</td><td colspan="2">List all the files and directories in your home directory, including hidden ones.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>ls -R (all files in the subdirectories home)
ls -a (hidden files)</td><td>chau@secr2043:~$ ls -R
.:
example.txt share

./share:
textfile01.txt
chau@secr2043:~$ ls -a
. .. .bash_logout .bashrc .cache .profile .ssh example.txt share</td></tr>
<tr><td rowspan="3">3.</td><td colspan="2">Create a new directory named "os_lab" in your home directory.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>Mkdir os_lab
ls</td><td>combined.txt file2.txt filelist.txt os_lab workspace
example.txt file3.txt linux-lab share</td></tr>
<tr><td rowspan="3">4.</td><td colspan="2">Change your current working directory to "os_lab".</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>cd os_lab
pwd</td><td>/home/chau/os_lab</td></tr>
<tr><td rowspan="2">5.</td><td colspan="2">Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
</table>

	touch hello.txt echo “Hello, world” > hello.txt	No output
6.	Display the content of "hello.txt".	
	Command	**Output**
	cat hello.txt	Hello,world
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.	
	Command	**Output**
	cp hello.txt hello_copy.txt	no output
8.	Rename "hello_copy.txt" to "hello_bak.txt".	
	Command	**Output**
	mv hello_copy.txt hello_bak.txt	No output
9.	Delete "hello.txt".	
	Command	**Output**
	rm hello.txt	No output
10.	Display your home directory with additional information, together with subdirectories contents.	
	Command	**Output**

<table>
<tr>
<td></td>
<td>ls -alhR ~</td>
<td><pre>chau@secr2043:~/os_lab$ ls -alhR ~
/home/chau:
total 60K
drwxr-x--- 8 chau chau 4.0K Apr 22 07:04 .
drwxr-xr-x 6 root root 4.0K Apr 16 06:16 ..
-rw-r--r-- 1 chau chau 220 Apr 16 06:16 .bash_logout
-rw-r--r-- 1 chau chau 3.7K Apr 16 06:16 .bashrc
drwx------ 2 chau chau 4.0K Apr 16 07:52 .cache
-rw-r--r-- 1 chau chau 807 Apr 16 06:16 .profile
drwx------ 2 chau chau 4.0K Apr 16 07:53 .ssh
-rw-r--r-- 1 chau chau 158 Apr 22 02:29 combined.txt
-rw-r--r-- 1 chau chau   0 Apr 22 00:21 example.txt
-rw-r--r-- 1 chau chau   79 Apr 22 02:25 file2.txt
-rw-r--r-- 1 chau chau   79 Apr 22 02:28 file3.txt
-rw-r--r-- 1 chau chau 316 Apr 22 02:26 filelist.txt
drwxr-xr-x 2 chau chau 4.0K Apr 22 02:22 linux-lab
drwxr-xr-x 2 chau chau 4.0K Apr 22 07:10 os_lab
drwxr-xr-x 2 chau chau 4.0K Apr 22 00:21 share
drwxr-xr-x 2 chau chau 4.0K Apr 22 02:20 workspace

/home/chau/.cache:
total 8.0K
drwx------ 2 chau chau 4.0K Apr 16 07:52 .
drwxr-x--- 8 chau chau 4.0K Apr 22 07:04 ..
-rw-r--r-- 1 chau chau   0 Apr 16 07:52 motd.legal-displayed

/home/chau/.ssh:
total 16K
drwx------ 2 chau chau 4.0K Apr 16 07:53 .
drwxr-x--- 8 chau chau 4.0K Apr 22 07:04 ..
-rw------- 1 chau chau 978 Apr 16 07:53 known_hosts
-rw-r--r-- 1 chau chau 142 Apr 16 07:53 known_hosts.old

/home/chau/linux-lab:
total 8.0K
drwxr-xr-x 2 chau chau 4.0K Apr 22 02:22 .
drwxr-x--- 8 chau chau 4.0K Apr 22 07:04 ..

/home/chau/os_lab:
total 12K
drwxr-xr-x 2 chau chau 4.0K Apr 22 07:10 .
drwxr-x--- 8 chau chau 4.0K Apr 22 07:04 ..
-rw-r--r-- 1 chau chau   13 Apr 22 07:10 hallo_bak.txt

/home/chau/share:
total 8.0K
drwxr-xr-x 2 chau chau 4.0K Apr 22 00:21 .
drwxr-x--- 8 chau chau 4.0K Apr 22 07:04 ..
-rw-r--r-- 1 chau chau   0 Apr 22 00:21 textfile01.txt

/home/chau/workspace:
total 8.0K
drwxr-xr-x 2 chau chau 4.0K Apr 22 02:20 .
drwxr-x--- 8 chau chau 4.0K Apr 22 07:04 ..</pre></td>
</tr>
<tr>
<td></td>
<td>Total Mark:</td>
<td></td>
</tr>
</table>

ACTIVITY 2 : **Text File Manipulation**

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

For each question, write down the command and the output (if any) in the answer sheet.

1. Create an empty file named "hello.txt" in the current working directory.

Command	Output
touch hello.txt	No output

2. Write the text "Hello World" to "hello.txt" using echo.

Command	Output
echo “Hello, world” > hello.txt	No output

3. Append the text "This is a test" to "hello.txt" using echo.

Command	Output
echo “This is a test” >> hello.txt	No output

4. Display the contents of "hello.txt" using cat.

Command	Output
cat hello.txt	chau@secr2043:~/os_lab$ echo "Hello, world" > hello.txt chau@secr2043:~/os_lab$ echo "This is a test" >> hello.txt chau@secr2043:~/os_lab$ cat hello.txt Hello, world This is a test

5. Create another file named "bye.txt" with the text "Goodbye World" using echo.

Command	Output
echo “Goodbye World” > bye.txt	No output

6. Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.

Command	Output
cat hello.txt bye.txt	chau@secr2043:~/os_lab$ echo "Goodbye World" > bye.txt chau@secr2043:~/os_lab$ cat hello.txt bye.txt Hello, world This is a test Goodbye World

7. Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.

Command	Output

	cat hello.txt bye.txt > greeting.txt	No output
8.	Display the contents of "greeting.txt" using cat.	
	Command	**Output**
	cat greeting.txt	chau@secr2043:~/os_lab$ cat hello.txt bye.txt > greeting.txt chau@secr2043:~/os_lab$ cat greeting.txt Hello, world This is a test Goodbye World
9.	Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.	
	Command	**Output**
	nano greeting.txt GNU nano 7.2 greeting.txt * Hello, Chau This is a test Goodbye World	No output
10.	Display the modified contents of "greeting.txt" using cat.	
	Command	**Output**
	cat greeting.txt	chau@secr2043:~/os_lab$ cat greeting.txt Hello, Chau This is a test Goodbye World
	Total Mark:	

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc.
For each question, write down the command and the output (if any) in the answer sheet.

1. Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".

Command	Output
nano hello.c	No output

```
  GNU nano 7.2                                hello.c
#include <stdio.h>
int main() {
        printf("Hello, World!\n");
        return 0;
}
```

2. Compile the program "hello.c" using gcc and generate an executable file named "hello".

Command	Output
gcc hello.c -o hello	No output

3. Run the executable file "hello" and display its output.

Command	Output
./hello	(see below)

```
chau@secr2043:~/os_lab/workspace$ ./hello
Hello, World!
```

4. Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".

Command	Output
nano sum.c	No output

```
  GNU nano 7.2                              sum.c *
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[]) {
    if (argc != 3) {
        printf("Usage: %s <num1> <num2>\n", argv[0]);
        return 1;
    }

    int num1 = atoi(argv[1]);
    int num2 = atoi(argv[2]);
    int sum = num1 + num2;

    printf("Sum: %d\n", sum);
    return 0;
}
```

5. Compile the program "sum.c" using gcc and generate an executable file named "sum".

Command	Output
gcc sum.c -o sum	No output

6. Run the executable file "sum" with 10 and 20 as arguments and display its output.

Command	Output
./sum 10 20	(see below)

```
chau@secr2043:~/os_lab/workspace$ ./sum 10 20
Sum: 30
```

<table>
<tr><td rowspan="3">7.</td><td colspan="2">Run the executable file "sum" with -5 and 15 as arguments and display its output.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>./sum -5 15</td><td><pre>chau@secr2043:~/os_lab/workspace$ ./sum -5 15
Sum: 10</pre></td></tr>
<tr><td rowspan="3">8.</td><td colspan="2">Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>nano echo.c<pre> GNU nano 7.2   echo.c *
#include <stdio.h>

int main() {
    char line[256];

    printf("Enter a line of text: ");
    fgets(line, sizeof(line), stdin);

    printf("You entered: %s", line);
    return 0;
}</pre></td><td>No output</td></tr>
<tr><td rowspan="3">9.</td><td colspan="2">Compile the program "echo.c" using gcc and generate an executable file named "echo".</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>gcc echo.c -o echo</td><td>No output</td></tr>
<tr><td rowspan="3">10.</td><td colspan="2">Run the executable file "echo", type "Hello World" as input and display its output.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>./echo
Hello World</td><td><pre>chau@secr2043:~/os_lab/workspace$ ./echo
Enter a line of text: Hello World
You entered: Hello World</pre></td></tr>
<tr><td></td><td>Total Mark:</td><td></td></tr>
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